CHAPTER A

The symmetric Laplacian in coordinates

The universal null construction depends on an exact differential operator. This appendix derives that operator from the Euclidean metric, rather than treating its power-sum formulas as a black box. It also explains the Newton reductions used by the implementation and gives two low-degree harmonic checks. All polynomial identities are on the hyperplane $H = \{v \in \mathbb{R}^5 : \sum_i v_i = 0\}$, with its induced Euclidean metric. The modularity and valence contracts remain the ones stated in Chapters **??** and **??**.

Write $s_m = \sum_{i=0}^{4} v_i^m$, with $s_0 = 5$ and $s_1 = 0$, and let $D = \prod_{i<j}(v_i - v_j)$. We distinguish the ordinary Laplacian Δ_H from the conjugated operator $\mathcal{L}$ defined by $\Delta_H(DQ) = D\mathcal{L}Q$. The factor D changes radial coefficients substantially. Confusing the two operators would invalidate the harmonicity test.

A.1 Two descriptions of the same metric

The orthogonal projection from $\mathbb{R}^5$ to H is

$$P = I_5 - \frac{1}{5}\mathbf{1}\mathbf{1}^T, \tag{A.1}$$

where $\mathbf{1} = (1,1,1,1,1)^T$. Indeed, subtracting the coordinate mean leaves a zero coordinate sum. The matrix is symmetric, satisfies $P^2 = P$, and has image H. For an ambient polynomial F, its gradient along H is $P\nabla F$. Its Laplacian along H is

$$\Delta_H(F|_H) = \left(\sum_{i=0}^{4} \partial_i^2 F - \frac{1}{5}\left(\sum_{i=0}^{4} \partial_i\right)^2 F\right)\Big|_H. \tag{A.2}$$

Only tangential derivatives enter. Thus this expression is independent of the polynomial extension away from H.

For direct symbolic computation, use four coordinates $x_0, \ldots, x_3$ and substitute

$$v = (x_0, x_1, x_2, x_3, -x_0 - x_1 - x_2 - x_3).$$

The coordinate tangent vectors are $b_a = e_a - e_4$. Their Gram matrix and its inverse are

$$G = I_4 + \mathbf{1}\mathbf{1}^T, \qquad G^{-1} = I_4 - \frac{1}{5}\mathbf{1}\mathbf{1}^T, \tag{A.3}$$

now with a four-entry vector $\mathbf{1}$. Matrix multiplication verifies the inverse. The factor $1/5$ remains because the missing fifth coordinate is determined by the four chosen coordinates.

The chart Laplacian is consequently

$$\Delta_H = \sum_{a,b=0}^{3} \left(\delta_{ab} - \frac{1}{5}\right) \frac{\partial^2}{\partial x_a \partial x_b}. \tag{A.4}$$

The coordinates x_a are not orthonormal. Using only the four diagonal second derivatives would use a different metric. As a normalization check, $s_2 = x^T G x$, so its Hessian is $2G$ and

$$\Delta_H s_2 = \operatorname{tr}(G^{-1} 2G) = 8.$$

The rank of the Euclidean space is four, giving twice that rank. Later we obtain $\mathcal{L} s_2 = 48$, an intentionally different number.

A.2 Conjugating by the Vandermonde

The Vandermonde is alternating under coordinate permutations. Since Δ_H commutes with those permutations, $\Delta_H D$ is alternating of degree eight. Every alternating polynomial vanishes on each hyperplane $v_i = v_j$. In the polynomial ring of H, the ten corresponding linear factors are distinct and divide it. Their product has degree ten. Therefore $\Delta_H D = 0$.

On the open set where $D \neq 0$, the product rule gives

$$\frac{\Delta_H(DQ)}{D} = \Delta_H Q + 2\nabla_H \log D \cdot \nabla_H Q. \tag{A.5}$$

Here $\nabla \log D$ denotes the rational logarithmic derivative $\nabla D / D$. It does not require choosing a complex logarithm or assuming that D is positive. For symmetric Q, the apparent poles cancel and the right side extends to a polynomial on all of H.

There is a second explanation of the extension. The polynomial $\Delta_H(DQ)$ is alternating, so D divides it. Its quotient is symmetric. Thus the operator $\mathcal{L}$ is well defined on symmetric polynomials even at coordinate collisions. The packet shell itself has no collisions because its residues are distinct. The polynomial extension is nevertheless needed when constructing and comparing symbolic identities.

Translation invariance of D gives

$$\sum_i \frac{\partial_i D}{D} = 0, \qquad \frac{\partial_i D}{D} = \sum_{j \neq i} \frac{1}{v_i - v_j}.$$

Its ambient logarithmic gradient is already tangential. This removes a potential extra projection term in the derivation below.

A.3 Deriving the power-sum operator

For $m \geq 2$, differentiation and projection give

$$(\nabla_H s_m)_i = m \left(v_i^{m-1} - \frac{1}{5} s_{m-1} \right), \tag{A.6}$$

$$\Delta_H s_m = \frac{4}{5} m(m-1) s_{m-2}. \tag{A.7}$$

In the logarithmic term, the coordinate-mean subtraction vanishes because the logarithmic gradient has zero sum. Pair the ordered indices i, j and j, i to obtain

$$2\nabla_H \log D \cdot \nabla_H s_m = 2m \sum_{i<j} \frac{v_i^{m-1} - v_j^{m-1}}{v_i - v_j}.$$

The divided difference is the finite polynomial

$$\frac{v_i^{m-1} - v_j^{m-1}}{v_i - v_j} = \sum_{a=0}^{m-2} v_i^a v_j^{m-2-a}.$$

The sum over a is symmetric in the two coordinates. Hence twice the sum over $i < j$ equals the sum over $i \neq j$. For each a, that ordered sum is $s_a s_{m-2-a} - s_{m-2}$. There are $m-1$ values of a. We obtain

$$2\nabla_H \log D \cdot \nabla_H s_m = m \sum_{a=0}^{m-2} s_a s_{m-2-a} - m(m-1)s_{m-2}.$$

Adding (A.7) proves

$$\boxed{\mathcal{L}s_m = m \sum_{a=0}^{m-2} s_a s_{m-2-a} - \frac{m(m-1)}{5} s_{m-2}.} \tag{A.8}$$

This formula is asserted for $m \geq 2$. The boundary values are separately $\mathcal{L}s_0 = 0$ and $\mathcal{L}s_1 = 0$, since s_0 is constant and s_1 vanishes on H. No negative-index power sum is introduced.

The first values are

$$\mathcal{L}s_2 = 48, \qquad \mathcal{L}s_3 = 0, \qquad \mathcal{L}s_4 = \frac{188}{5} s_2, \qquad \mathcal{L}s_5 = 46 s_3. \tag{A.9}$$

For example, at $m = 2$ the first term is $2s_0^2 = 50$, while the subtraction is $2s_0/5 = 2$. The additional 40 compared with $\Delta_H s_2 = 8$ comes from differentiating the degree-ten factor D.

A.4 The chain-rule metric and polynomial assembly

The same projected gradients give

$$\Gamma_{mn} := \nabla_H s_m \cdot \nabla_H s_n = mn \left(s_{m+n-2} - \frac{1}{5} s_{m-1} s_{n-1} \right). \tag{A.10}$$

Expanding the product in (A.6) gives the first power sum once and the coordinate-mean product with coefficient $-1/5$. This identity includes $m, n \geq 1$, with the evident zero tangential gradient for s_1.

If $Q = F(s_2, s_3, s_4, s_5)$, the ordinary second-order chain rule and (A.5) yield

$$\mathcal{L}Q = \sum_{m=2}^{5} \frac{\partial F}{\partial s_m} \mathcal{L}s_m + \sum_{m,n=2}^{5} \frac{\partial^2 F}{\partial s_m \partial s_n} \Gamma_{mn}. \tag{A.11}$$

The second sum runs over ordered pairs. Thus a mixed derivative contributes twice when $m \neq n$. If one instead sums over $m \leq n$, the off-diagonal terms require a factor two. These are two equivalent conventions, but their factors must not be mixed.

Useful instances are

$$\mathcal{L}(s_2^2) = 104 s_2, \tag{A.12}$$

$$\mathcal{L}(s_2^3) = 168 s_2^2, \tag{A.13}$$

$$\mathcal{L}(s_2 s_4) = 64 s_4 + \frac{188}{5} s_2^2, \tag{A.14}$$

$$\mathcal{L}(s_3^2) = 18 s_4 - \frac{18}{5} s_2^2. \tag{A.15}$$

For the first identity, $2s_2\mathcal{L}s_2 + 2\Gamma_{22} = 96s_2 + 8s_2$. For the mixed product, $s_4\mathcal{L}s_2 + s_2\mathcal{L}s_4 + 2\Gamma_{24}$ gives the displayed coefficients. Each calculation is a check of the ordered-pair convention.

The metric formula may produce s_6, s_7, s_8. They are not new independent variables. The invariant ring on H is generated by s_2, s_3, s_4, s_5, and Newton's identities reduce higher indices exactly.

A.5 Newton reductions with zero first power sum

Let e_j be the elementary symmetric polynomials in the five coordinates. Their characteristic polynomial is

$$\prod_i (T - v_i) = T^5 + e_2T^3 - e_3T^2 + e_4T - e_5,$$

because $e_1 = s_1 = 0$. The first Newton identities give

$$e_2 = -\frac{1}{2}s_2, \qquad e_3 = \frac{1}{3}s_3, \tag{A.16}$$

$$e_4 = \frac{1}{8}s_2^2 - \frac{1}{4}s_4, \qquad e_5 = \frac{1}{5}s_5 - \frac{1}{6}s_2s_3. \tag{A.17}$$

For instance, the degree-five identity is $s_5 + e_2s_3 - e_3s_2 - 5e_5 = 0$. The two middle terms combine to $-5s_2s_3/6$, fixing the sign in e_5.

Every coordinate satisfies its characteristic polynomial. Multiplying that equality by v_i^{k-5} and summing, for $k \geq 6$, gives

$$s_k = -e_2s_{k-2} + e_3s_{k-3} - e_4s_{k-4} + e_5s_{k-5}. \tag{A.18}$$

This recurrence uses only lower indices. Substitution at $k = 6, 7$ yields

$$s_6 = \frac{3}{4}s_2s_4 + \frac{1}{3}s_3^2 - \frac{1}{8}s_2^3, \tag{A.19}$$

$$s_7 = \frac{7}{10}s_2s_5 + \frac{7}{12}s_3s_4 - \frac{7}{24}s_2^2s_3. \tag{A.20}$$

The same rule gives

$$s_8 = \frac{1}{4}s_4^2 + \frac{1}{4}s_2^2s_4 - \frac{1}{16}s_2^4 + \frac{8}{15}s_3s_5. \tag{A.21}$$

Each term has the required invariant degree. This supplies a useful local check before substitution into a larger operator expression.

At the first packet $v = (0, 1, 2, -2, -1)$, the first power sums are $s_2 = 10$, $s_3 = 0$, $s_4 = 34$, $s_5 = 0$. Formula (A.19) gives $255 - 125 = 130$, matching the direct sum of sixth powers. Formula (A.20) gives zero, matching cancellation of the opposite coordinates. Formula (A.21) gives 514. These examples fix normalizations. The polynomial recurrences, rather than these examples, establish the identities on every vector in H.

A.6 Radial factors and harmonic projection

Let h_ℓ be homogeneous harmonic of degree ℓ on the rank-four space H. Euler's identity is $v \cdot \nabla_H h_\ell = \ell h_\ell$. The product rule for the radial factor s_2^j gives

$$\Delta_H(s_2^j h_\ell) = 4j(\ell + j + 1)s_2^{j-1}h_\ell, \qquad j \geq 1. \tag{A.22}$$

At $j = 0$ the left side is zero. At $j = 1, \ell = 0$ the coefficient is 8, agreeing with $\Delta_H s_2$. At $j = 1, \ell = 10$ it is 48, agreeing with $\Delta_H(s_2 D) = 48D$.

For a homogeneous polynomial P of degree M, a concrete top-degree harmonic projection in dimension four is

$$\mathcal{H}_M(P) = \sum_{j=0}^{\lfloor M/2 \rfloor} \frac{(-1)^j (M-j)!}{4^j j! M!} s_2^j \Delta_H^j P. \tag{A.23}$$

For $M = 0$ or 1 the sum contains only $j = 0$. In general, if a_j denotes its coefficient, the product rule gives the cancellation condition

$$a_j + 4(j+1)(M-j)a_{j+1} = 0.$$

The factorial coefficients satisfy this condition. The last possible iterated Laplacian has no further nonzero derivative, so the resulting polynomial is harmonic. Its difference from P is divisible by s_2.

Repeating the construction on that lower-degree quotient produces the harmonic decomposition

$$P = \sum_{j=0}^{\lfloor M/2 \rfloor} s_2^j h_{M-2j}. \tag{A.24}$$

It is unique. If such a sum were zero, apply the Laplacian as many times as the largest occupied radial exponent. All smaller-exponent terms vanish. Formula (A.22) leaves a nonzero scalar times the last harmonic polynomial, so that polynomial must be zero. Descending removes every term.

When $P = DQ$ is alternating, each projected harmonic component is alternating because the Laplacian and multiplication by s_2 commute with permutations. Each nonzero component is divisible by D and has degree at least ten. Thus for a correction of homogeneous degree d, only $j \le d/2$ can occur. This is the degree bound used in the quasimodular argument for nonhomogeneous content conservation.

Harmonic projection is not preservation of shell values. The elementary counterexample is $P = s_2 D$, of degree 12. Since $\Delta_H P = 48D$ and the next Laplacian is zero,

$$\mathcal{H}_{12}(s_2 D) = s_2 D - \frac{1}{48} s_2 (48D) = 0.$$

Yet on $\mathcal{V}_n$ the original polynomial equals $10nD$, which is nonzero at every packet. The operation removes radial information. It cannot silently turn a shell interpolant into a universally null polynomial with the same values.

A.7 Two explicit harmonic moment checks

The degree-four correction

$$K_4 = 130 s_4 - 47 s_2^2$$

has

$$\mathcal{L} K_4 = 130 \frac{188}{5} s_2 - 47(104 s_2) = 0.$$

Thus DK_4 is harmonic. At the first packet, $W = 1$ and $K_4 = 130 \cdot 34 - 47 \cdot 100 = -280$. Its theta form has weight 16, whose cusp space is spanned by ΔE_4 with first coefficient

one. Consequently the theta form is $-280\Delta E_4$, giving the moment identity already stated in Chapter **??**.

Likewise, set

$$K_6 = 90s_2s_4 - 27s_2^3 - 320s_3^2.$$

Using (A.13)–(A.15), its operator image is

$$\begin{aligned}\mathcal{L}K_6 &= (5760 - 5760)s_4 \\ &\quad + (3384 - 4536 + 1152)s_2^2 = 0.\end{aligned}$$

At the first packet its value is $90 \cdot 10 \cdot 34 - 27 \cdot 1000 = 3600$. The weight-18 cusp space is spanned by ΔE_6, again normalized with first coefficient one. Its theta form is therefore $3600\Delta E_6$.

These checks have different roles from the null certificates. Both polynomials are harmonic, but neither is universally null: their first coefficients are nonzero. Harmonicity supplies a modular-form space. The initial coefficients then identify the form or, when enough initial coefficients vanish, force it to be zero by valence. Skipping that second step would misidentify a nonzero moment as a permitted null correction.

The retained programs organize these same operations using sparse rational polynomials. A direct four-coordinate implementation can instead substitute $v_4 = -\sum_{a=0}^{3} x_a$ into DQ and apply (A.4). Agreement between these organizations checks the symbolic construction through different algebraic representations. The universal justification remains the metric calculation, the exact polynomial identities and the stated harmonic theta and valence contracts. None of those statements alone excludes the unresolved zero prime parameter of Δ.